Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/05/2009
PI Name:	Cole, Kathleen S Ph.D.		
Project Title:	Effects of Prolonged Exposure to a Ground-Based Analog for Microgravity During Embryonic Development on Skeletal Morphogenesis, Cephalization, Post-Hatch Behavior and Subsequent Reproductive Development in the Isogenic Fish, Rivulus marmoratus		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	95622	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	02/15/2004	**End Date:**	02/14/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 2/14/2008 per A. Chu (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC58G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Rivulus marmoratus is a hermaphroditic atheriniform teleost fish unique among vertebrates in its ability to internally self-fertilize and produce isogeneic offspring. In preliminary trials supported by BOR DGAP and NASA-LaSpace programs and run at the Johnson Space Center, R. marmoratus embryos have been successfully maintained in a simulated microgravity environment (as provided by a ground-based rotating cell culture system (RCCS)) for up to four weeks prior to hatching and for up to 7 days following hatching. Newly hatched juveniles maintained in an RCCS environment exhibit pronounced body curvatures, limited mobility, and deficits in tracking and capture of live prey. In addition, pronounced cranial depression suggests possible deficiencies in cephalization. None of these features were observed in control Rivulus. The proposed research addresses four questions related to the Developmental Biology FSB program element. The first goal is to determine if abnormalities of the axial skeleton observed in R. marmoratus following RCCS exposure during embryonic development are due to abnormalities associated with either the initial formation of dense connective tissue (cartilage and dermal bone) or the subsequent organization of skeletal elements into a articulating infrastructure (skeletogenesis). The second goal is to determine if additional observed abnormalities of the cranial skeleton and severe deficits in post-hatch behavior among RCCS-reared R. marmoratus are associated with abnormalities in brain structure caused by alterations in brain morphogenesis. The third goal is to determine if exposure to an RCCS- simulated microgravity environment during embryonic development of R. marmoratus alters the normal development of ovarian and testicular tissue within the ovotestis during gonadogenesis, or alters the timing and trajectory of maturation and subsequent reproductive function and the capability to produce viable offspring. The fourth goal is to test the hypothesis that, as demonstrated in other model organisms, early developmental stages in R. marmoratus are more vulnerable to the deleterious effects of exposure to RCCS-simulated microgravity on morphogenesis than exposure during later stages of development.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Not Available
Task Progress:	Previous work by the PI and Co-Is demonstrated that exposure to modeled microgravity (provided by the Rotating Cell Culture System©, or RCCS, by Synthecon) of embryonic Rivulus marmoratus (Teleostei) resulted in profound developmental abnormalities. These included extreme torsion of the axial skeleton, depression of the post-ocular cranial vault, and severely impaired sensory system responses (lack of visual tracking, feeding response or responsivity to mechanical stimulation) post-hatch. All fish experiencing an RCCS environment throughout their ontogeny exhibited abnormalities and none survived for more than a week post-hatch. The initial goals of the project were to: (1) characterize the topology, composition and placement of skeletal elements associated with axial torsion; (2) determine if abnormalities of the cranial skeleton and severe deficiencies in post-hatch behavior were accompanied by abnormalities in brain structure and/or brain morphogenesis; (3) determine if prolonged exposure to modeled microgravity during development altered gonadogenesis and/or and the capability to produce viable offspring; (4) document and evaluate the effects of RCCS rearing conditions on post-hatch locomotion, sensory responsivity and feeding behavior; and (5) test the hypothesis that early exposure to modeled microgravity causes more severe developmental abnormalities than exposure during later developmental stages. However, lack of survivorship post-hatch among experimental Rivulus precluded investigations associated with Goals 1, 3, and 4. Therefore, work carried out for this research project focused on Goals 2 and 5, related to induced developmental abnormalities evident at the end of the embryonic stage of development. Summary of Project Accomplishments Based on the findings of this project, Rivulus marmoratus is an excellent vertebrate model for evaluating the effects of prolonged exposure to modeled microgravity during embryonic development using the Synthecon Rotating Cell Culture System©. Embryo survivorship was high for both experimental and control fish, although developmental abnormalities associated with modeled microgravity exposure were sufficiently severe as to limit post-hatch survival to the end of the endogenous (i.e., yolk-sourced nutrients) phase. Therefore, persistence of morphogenic abnormalities in post-hatch individuals, including deviations from normal skeletal development, gonad development, mobility and behavior could not be determined. Results from this project have been presented at three conferences, including the American Society of Ichthyologists and Herpetologists and the International Congress of Developmental Morphologists in 2004, and the NASA Cell Conference in Galveston in 2005. Two manuscripts are in preparation for submission for journal review.
Bibliography Type:	Description: (Last Updated: 02/27/2004)